Surface Ion Trap Optimization

Geometry, stability and fabrication

A constrained numerical study of a single calcium-ion surface trap

153.643 meV • 49.82812 MHz • 100 V peak

Source: Final notebook: parameters and saved outputs

DESIGN OBJECTIVE

Objective: maximize depth within the constraints

Requirement	Allowed value
Ion height / radial frequency	50–100 μm / ≥1 MHz
Mathieu / peak voltage	q ≤0.4 / ≤100 V
RF width / gap	20–200 μm / 5–20 μm
RF drive	10–60 MHz

Expanded scope: center width and layer thicknesses are design variables.

Source: Starter notebook: constraints and RF model

Source: Final notebook: parameters and saved outputs

Development: hypothesis, failure, correction

- Initial hypothesis: lower ion height and RF frequency to strengthen trapping.
- Failed trials showed that lowering frequency could violate the Mathieu limit.
- Correct the electrode-separation ratio and tune frequency near q =0.4.

Recorded progress: ~150.5 meV milestone → latest saved result 153.643 meV.

Source: Final notebook: development notes and saved score

Source: Nizamani & Hensinger, 2012 • doi:10.1007/s00340-011-4803-x

Why the selected effective ratio is about 6.46

Depth □ V^2/f^2 q □ V G/f^2

- 3.68 refers to RF width / RF separation—not the fabrication gap.
- The constrained search varies height and frequency, changing the optimum.
- (200 + 5)/(26.71361 + 5) ≈ 6.46 in the effective-width model.

Source: Nizamani & Hensinger, 2012 • doi:10.1007/s00340-011-4803-x

Source: Project constrained geometry model: optimize_trap.py

Symmetric rails led the modeled depth search

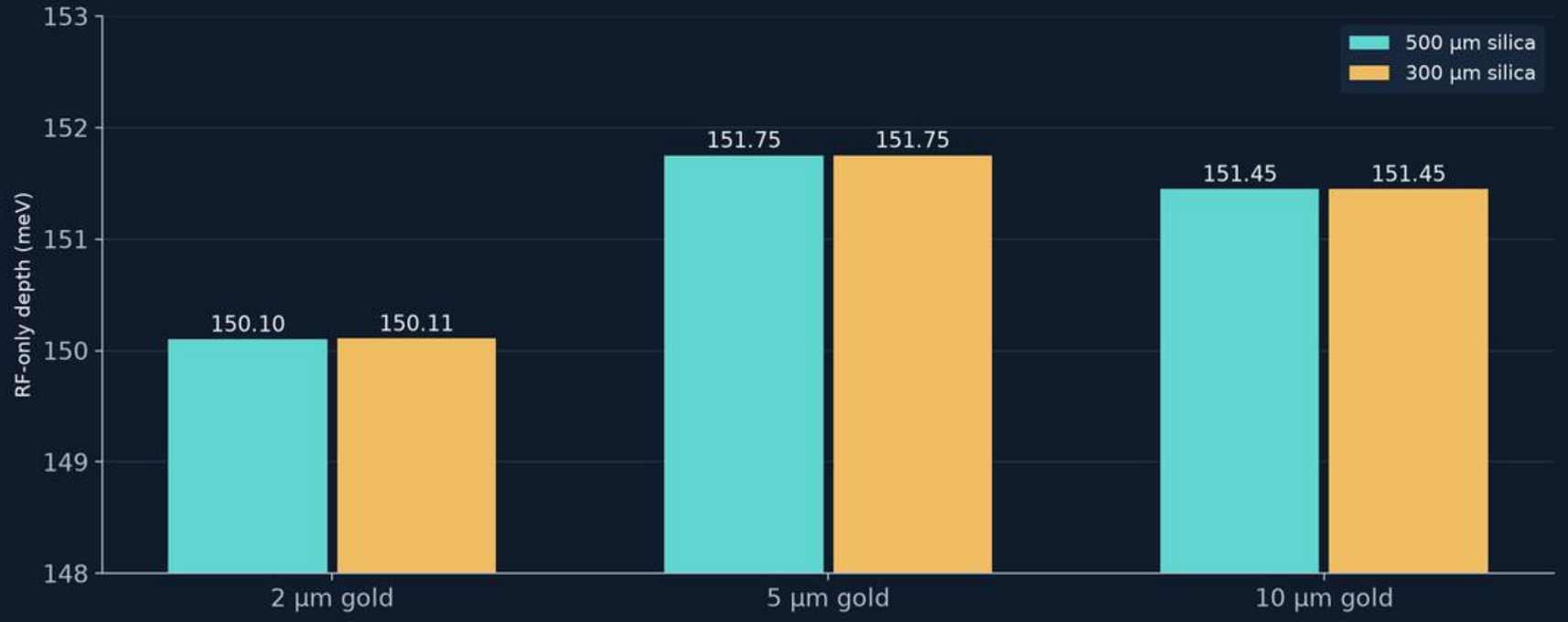
Fixed 60 µm center	Predicted depth
Equal rails: 200 / 200 µm	144.29 meV
Independent widths: ≈200 / 200 µm	144.29 meV
Unequal example: 200 / 100 µm	134.20 meV

Asymmetry may help mode orientation; it did not improve this depth objective.

Source: Project analytic results: trap_optimization_results.json

Source: Allcock et al., 2010 • doi:10.1088/1367-2630/12/5/053026

Thickness selection: 5 µm gold, 500 µm silica



Retain the 500 µm wafer: thinning offered negligible modeled depth benefit.

Source: Project thickness screen: thickness_screen_results.json

Source: Britton, 2008 • plated-gold trap fabrication

Gold-coated copper: a loss-reduction option

~23% lower estimated electrode resistance

- Candidate stack: 4.8 µm copper +0.2 µm gold.
- Potential benefit: less conductor loss at the same electrode voltage.
- No demonstrated trap-depth increase at the same 100 V amplitude.

Source: Tao et al., 2020 • arXiv:2002.11470

Source: Matula, 1979 • bulk resistivity reference data

FINAL RESULT

Final notebook: design and saved performance

Design	Value
RF width / gap / center	200 /5 /26.71361 µm
Gold / fused-silica thickness	5 /500 µm
Voltage / RF frequency	100 V peak /49.82812 MHz

Saved RF-only depth

153.643 meV

Height above metal

58 µm

Mathieu parameter

0.39999986

Source: Final notebook: parameters and saved outputs

Conclusion

The design passes all coded constraints at the current simulation resolution.

- Symmetric RF rails and a constrained frequency search produced the selected design.
- Several-micrometre gold is a practical fabrication candidate.
- The result is an RF-only prediction; device performance remains to be validated.

Source: Final notebook: score and model assumptions

Source: Schmied, 2010 • finite-electrode modeling, doi:10.1088/1367-2630/12/2/023038